# Assignment 3: Association Rule Mining for Market Basket Analysis

Data Analytics 1

Release: October 3, 2026
Deadline: October 14, 2026 (11:55 pm)

### Marking Distribution (Total 100 Marks)

- **Part A: Data Preprocessing** 15 Marks
- **Part B: Association Rule Mining** 35 Marks
- **Part C: Recommendation and Evaluation** 35 Marks
- **Report and Code Quality** 15 Marks

## 1 Objective

The objective of this assignment is to perform market basket analysis to discover associations between movies watched and liked together. You will build a system that can suggest movies to a user based on the movies already present in their watchlist using association rules.

## 2 Dataset Description

You will use the MovieLens 100K dataset from GroupLens Research. The dataset contains user rating activity across thousands of movies. We will primarily use the following files:

- `ratings.csv`: Contains rating records with fields: `userId`, `movieId`, `rating`, `timestamp`.
- `movies.csv`: Contains movie details with fields: `movieId`, `title`, `genres`.

## 3 Assignment Tasks

### Part A: Data Preprocessing [15 Marks]

1. **Data Filtering & Binarization (5 marks):**
   - Binarize ratings: Filter ratings in the MovieLens dataset to retain only items with $rating \geq 3.0$ (treating 3+ stars as a positive interaction or "liked movie").
   - Filter low-activity users: Remove users who have fewer than 5 liked movies to ensure meaningful basket evaluation.
2. **Form the Transactional Dataset (5 marks):** Group the filtered interactions by `userId` to create a list of transactions, where each transaction is a `userId` associated with the set of unique `movieId` values (or movie titles) liked by that user.
3. **Split the Data (5 marks):** Randomly divide your transactional dataset into a training set (80% of user transactions) and a test set (20% of user transactions).

**Part B: Association Rule Mining [35 Marks]**

1. **Mine for Rules (25 marks):** From your 80% training set, extract all association rules of the form $\{A\} \rightarrow Y$ where $A$ contains a single movie and $Y$ can contain one or more movies. You may use either the Apriori or FP-growth algorithm.

2. **Experiment with Thresholds (10 marks):** You must experiment with different values for minimum support (*minsup*) and minimum confidence (*minconf*) to find a combination that produces a meaningful number of rules. For example, you could try values for *minsup* in the range $[0.01, 0.1]$ and for *minconf* in the range $[0.1, 0.5]$. Plot suitable graphs to arrive at an optimum value and justify your final choice in the report.

**Part C: Recommendation and Evaluation [35 Marks]**

1. **Generate Rule Lists (5 marks):** Create two lists of the top 100 association rules found: one sorted by support and the other by confidence. Identify and present the rules that appear in both lists.

2. **Implement Evaluation Logic (15 marks):** For each transaction in your 20% test set with $> 1$ movie, split the items: use the first half of the movies as input and the second half as the ground truth to be predicted.

3. **Calculate and Plot Metrics (10 marks):** Compute the average precision and average recall. Generate plots showing how these metrics change as you vary the number of recommendations per user (from 1 to 10).

4. **Sample Analysis (5 marks):** Show the precision and recall graphs for a few sample users from your test set and briefly discuss their results.

## 4 Report and Code Quality [15 Marks]

Your submission will also be graded on the quality of your report and code.

- **Report (10 marks):** The report must be clear, well-structured, and provide justifications for your choices (e.g., algorithm, final *minsup*/*minconf*). It should include all plots and a thorough interpretation of your findings and results.

- **Code (5 marks):** The code should be well-commented, readable, and efficiently implemented.

## 5 Submission Format

Please submit a single zip folder named `<assignment3_teamId>.zip` containing:

- `<TeamId>_report.pdf`: Your report justifying your algorithm, data preprocessing steps, and results.

- `<TeamId>_recommender.ipynb`: Your fully commented code.

- `<TeamId>_top100RulesByConf.txt`: Your list of top 100 rules by confidence.

- `<TeamId>_top100RulesBySup.txt`: Your list of top 100 rules by support.